Fermium Hazard Mapper: flood risk mapping trained on radar-observed floods and tested on held-out areas

Md Rakib Hasan[a,b,*], Shoumik Zubyer[a]

[a] *Department of Soil, Water and Environment, University of Dhaka, Dhaka, 1000, Bangladesh*
[b] *Fermium Systems, Dhaka, 1207, Bangladesh*

Abstract

Flood susceptibility maps for Bangladesh are usually trained on a few hundred surveyed points and scored on a random share of the same points, so their accuracy says little about unmapped ground. Fermium Hazard Mapper is an open-source Python pipeline and web application that trains flood models on extents mapped from Sentinel-1 radar, scores every model on whole areas held out from training, and publishes risk for each mapped asset and administrative unit. Across 8 flood regions covering all 61 lowland districts and 114,569 assets, the default model reaches an AUC of 0.724 to 0.914 on held-out 10 km blocks, against 0.49 to 0.66 for the wetness index alone, and training on radar-observed floods beats training on terrain-threshold labels in every region. The software's contribution is a validation suite in which every comparison runs on identical held-out areas, reports its spread and reruns with one command, so another group can check any claim about a model or a data source. Results are served openly at https://fermium.systems/hazmapper.

Keywords: flood susceptibility, Sentinel-1, spatial cross-validation, machine learning, risk mapping, open-source software, Bangladesh

*Corresponding author.

Email addresses: rakibhhridoy@fermium.systems (Md Rakib Hasan), shoumik-2019617365@swe.du.ac.bd (Shoumik Zubyer)

URL: https://orcid.org/0009-0002-4007-7590 (Md Rakib Hasan), https://orcid.org/0009-0006-4085-1493 (Shoumik Zubyer)

Metadata

Nr.	Code metadata description	Metadata
C1	Current code version	v3.0
C2	Permanent link to code/repository used for this code version	https://github.com/rakibhhridoy/rimes-mapathon
C3	Permanent link to reproducible capsule	Archived release: https://doi.org/10.5281/zenodo.23108192
C4	Legal code license	MIT
C5	Code versioning system used	git
C6	Software code languages, tools and services used	Python 3.11; GeoPandas, Rasterio, GDAL, scikit-learn, PyTorch Geometric, PyKrige, PySAL; Google Earth Engine; FastAPI, SQLite, PMTiles and MapLibre GL
C7	Compilation requirements, operating environments and dependencies	Python 3.11 on Linux or macOS; minimum versions in requirements.txt and the tested versions in requirements-lock.txt; the GDAL command-line tools for the web build; a Google Earth Engine project only for the Sentinel-1 stage and the optional covariates
C8	Link to developer documentation/manual	https://github.com/rakibhhridoy/rimes-mapathon#readme
C9	Support email for questions	rakibhhridoy@fermium.systems

1. Motivation and significance

Bangladesh floods every monsoon, yet the susceptibility maps published for its flood regions rarely show how they would fare on ground they were not trained on. Studies in the Teesta basin, the Sylhet haor and the south-west coast report AUCs between 0.81 and 0.984 [1, 2, 3, 4], each from a random split of a few hundred to a few thousand points. A random split of spatially clustered points tests a model mostly on locations next to its training points, and block cross-validation, which holds out whole areas, exists to remove that advantage [5]. Where the two have been compared

the gap can be large, as in a forest-biomass model that explained more than half the variance under random validation and almost none under spatial validation [6]. The flood locations themselves vary in kind, from field surveys to single satellite scenes, so the same accuracy figure can measure agreement with quite different things.

A stricter design is now practical, because Sentinel-1 radar sees through monsoon cloud and has been processed into flood maps of high accuracy, including for the 2017 Bangladesh floods [7, 8], while OpenStreetMap records the roads, schools, hospitals, shelters and embankments that a flood would reach.

Neither the published studies nor the operational services reviewed in Section 1.1 join radar flood maps and mapped assets in one reproducible chain that trains on what the radar saw rather than on terrain thresholds, holds out whole areas when scoring, and puts the result in front of planners with its limits attached. Fermium Hazard Mapper is built to fill that gap by answering three questions for a configured region, namely where the ground is likely to flood, which mapped assets and which people stand on that ground, and how far those people are from the services that would help them cope. It then reports how well each answer holds up on areas held out from training, on floods that occurred after the training period, and against a model-free reference. This article describes that software, the evidence it reports for Bangladesh, and how it can be pointed at a new region.

1.1. Related software and studies

Machine-learning susceptibility models have grown richer in inputs and algorithms while keeping the same test. Recent studies in Iran combine up to 32 climatic, soil, land-use and terrain factors with random forests, boosting and maximum-entropy models and report AUCs near 0.9 [9, 10], one of them on a random 70/30 split of field-surveyed flood points, which is also the design of the Bangladesh studies above. Spatial validation is routine in species distribution modelling, where tools build spatially separated folds [11] and the area of applicability marks where the training data support a prediction at all [12]. Fermium Hazard Mapper brings that practice to flood susceptibility and adds the label source as a tested choice, comparing radar-observed and terrain-threshold labels on the same held-out blocks.

Existing services answer neighbouring questions. Google's Flood Hub forecasts river flooding days ahead from a global streamflow model [13], and the Copernicus Global Flood Monitoring service maps flood extents from

every incoming Sentinel-1 scene [14], but neither says which mapped assets stand on ground that floods again and again. The government's Multi-hazard Risk and Vulnerability Assessment publishes return-period scenario maps for Bangladesh [15], without asset-level scores or a held-out test that a user can rerun.

2. Software description

2.1. Software architecture

The system has three layers, a processing pipeline, a validation suite and two front ends, and they communicate only through files (Fig. 1). The pipeline is a Python package driven by one command-line interface, in which each stage reads the files the previous stage wrote into the region's data directory and writes its own. A region runs end to end with a single command,

```
python -m pipeline.cli -c configs/national/central.yaml run
```

and any stage can be rerun alone after a change to its settings. Because stages share nothing but files, every intermediate product can be opened in a GIS for inspection.

A region is defined by a YAML file that states only what differs from a shared base configuration and names that base with an extends key. The base holds the model settings, weights and thresholds, so a change to any of them reaches every region and results stay comparable. A region file carries its extent, either a bounding box or a list of districts, together with a country code, a projected coordinate system, data directories and, for validation, the flood events to map from radar, each with the dates of its window and the report those dates came from. A region defined by districts keeps only the assets, grid cells and validation points inside them, so the regions of a national partition never count the same asset twice.

The public site is the web application, which serves the outputs from an SQLite database and PMTiles vector tiles through a FastAPI service and a MapLibre GL client, so that every asset of every region is drawn at once and panning and filtering happen in the browser (Fig. 2). The build step also places every asset in its union, upazila, district and division, and sums each region's results up that hierarchy, so the map shows one administrative level at a time, chosen by zoom, and a click or a cascade of pickers narrows the assets and counters to one area. The official return-period scenarios of the government's Multi-hazard Risk and Vulnerability Assessment can be laid

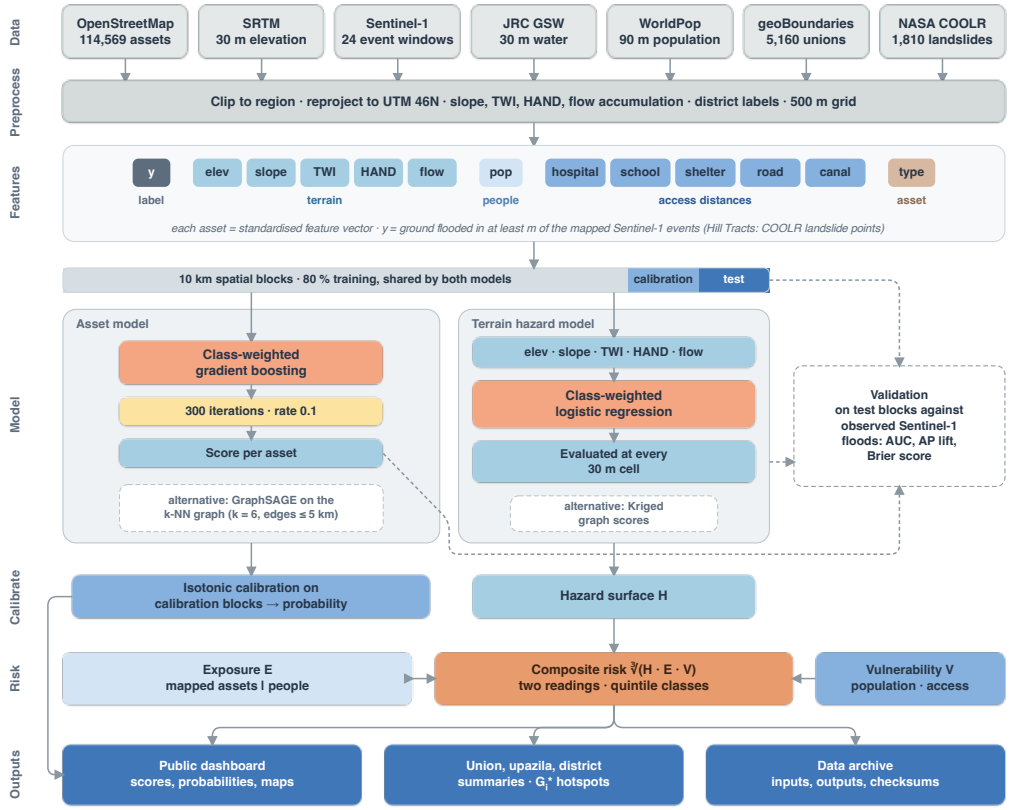


Figure 1: The processing chain, top to bottom, with the counts of the national run. Each asset enters the models as a vector of terrain, population, access-distance and asset-type features, with a label from the Sentinel-1 flood frequency. Every model trains on the same spatial blocks, and dashed connectors mark what the validation scores on the test blocks. In the landslide region a model fitted to a mapped inventory takes the place of the flood labels.

under the assets, drawn live from the server that publishes them, and the address of the page records the view, so a copied link or a downloaded image reproduces it. A Streamlit dashboard that reads the outputs directly remains for local use. Neither front end computes anything, and every value a visitor sees can therefore be traced to a file the pipeline wrote.

2.2. Software functionalities

The data stages listed in Table 1 read OpenStreetMap assets either from the Overpass API, in tiles that are cached and split when the server cannot answer, or from a national extract converted once into an indexed file. Assets are clipped to the region and to the country, so that an extent crossing a land border keeps only the home side. Slope, the topographic wetness index, height above nearest drainage and flow accumulation come from 30 m SRTM elevation, population from WorldPop, and administrative units from geoBoundaries, with divisions dissolved from their districts.

The Sentinel-1 stage maps each configured event in Google Earth Engine from ground-range-detected VV backscatter in interferometric wide-swath mode, by change detection against a dry-season baseline as UN-SPIDER recommends [16]. For each event it takes the per-pixel minimum over the flood window and the median over a dry-season window of the same year, and smooths both against speckle with a focal mean of 50 m radius. Water is marked where backscatter during the flood is below -16 dB and at least 3 dB lower than in the dry season. It then removes permanent water, where the JRC surface-water record [17] finds water at least half the time, slopes above 5°, where radar shadow mimics water, and patches under eight pixels. The event masks are averaged into a flood-frequency layer at 100 m, and a cell becomes a flood label when it was under water in a configured number of events, two in the North-west, Jamuna east, Sylhet haor and Central regions so that one season's flooded paddy does not count, and one in the other four. A download the service refuses is split into quarters.

Each asset enters the models as twelve features. Five describe the ground and how water moves over it, namely elevation, slope, height above nearest drainage, which measures how far water must rise to reach the asset, the topographic wetness index, which grows where a large upslope area drains through flat ground, and flow accumulation, which marks where runoff converges. The distance to the nearest mapped irrigation channel stands in for surface water, while population density, the distances to the nearest hospital,

Stage	What it does
download, ingest	fetch assets, elevation, population, boundaries and surface water, and clip them to the region
clip	drop assets across a land border from an asset file fetched earlier
preprocess	derive terrain rasters and write both label sources, radar-observed and terrain-threshold
sentinel1	map each configured flood event from radar and build the flood-frequency layer
features, graph	sample features at every asset and assign assets to 10 km spatial blocks
train	fit the configured asset model on training blocks and calibrate it on calibration blocks
krige, risk	build the terrain and Kriged hazard surfaces, exposure, vulnerability and composite risk, and summarise them by administrative unit
landslide	fit a landslide susceptibility model to a mapped inventory
validate, crosscheck	score models and surfaces against observed floods, and radar masks against an independent flood database
benchmark, sensitivity	compare models and label sources over repeated block assignments, run the temporal hold-out, and perturb the risk weights
run	run the whole chain for one region

Table 1: Pipeline stages, each a subcommand of the command-line interface.

school, flood shelter and road, and the asset type describe the settled landscape around an asset, and the importance of each is measured by shuffling it across the held-out assets. Every region is trained separately on its own events, so the flash floods of the haor, which begin in March, the riverine floods of the north-west and the surge of the south-west coast each teach their own model. A flood regime therefore enters the pipeline through its event list and label rule, with no change to the code.

The asset model is set in the configuration, with class-weighted gradient boosting as the default and a random forest, logistic regression and a two-layer GraphSAGE network [18] on a k-nearest-neighbour graph of assets available as alternatives. Gradient boosting runs 300 iterations at a learning

rate of 0.1, the random forest grows 300 trees with at least five assets per leaf, and the graph network has 64 hidden channels on each asset's six nearest neighbours and trains for up to 200 epochs with early stopping. Scores are calibrated by isotonic regression [19] fitted on blocks the model never trained on, with each calibration level shrunk towards the middle in proportion to how few points it holds, so that a level of three flooded points cannot return a certainty. Two hazard surfaces are written, a logistic model of each grid cell's own terrain and an ordinary Kriging of the asset scores, and the composite risk uses the one the configuration names. Composite risk is the geometric mean of hazard, exposure and vulnerability on a 500 m grid, the aggregation the INFORM index uses [20], and it is written twice, once for mapped infrastructure and once for population, with hotspots located by the Getis–Ord G_i^* statistic [21]. An experimental landslide module fits a logistic model of five terrain features to a mapped inventory, drawing background points only from the ground the inventory covers so that the model learns terrain rather than the mapping footprint [22]. In the Chittagong Hill Tracts, trained on 1,810 landslides mapped after one storm, it reaches an AUC of 0.729 on one assignment of held-out blocks, where landslides make up 16.7 % of points. It therefore shows where that storm triggered failures and is not a validated susceptibility map, and its results are not part of the evidence reported here.

The validation suite distinguishes the system from a mapping tool. Every model is scored on 10 km blocks held out from training, and the benchmark stage refits it over 20 block assignments so that each difference is reported with its spread. On identical features, assets and blocks, the same stage compares the configured models with one another and with the wetness index alone, compares training on radar labels with training on terrain-threshold labels, and trains on earlier floods to score against the latest year. The crosscheck stage compares the radar masks with the MODIS-based Global Flood Database [23], and the sensitivity stage recomputes the composite under alternative and randomly perturbed weights.

2.3. Safeguards against silent failure

A geospatial pipeline can fail without raising an error, as when a raster stored in a projected system is sampled with longitude and latitude and returns its nodata value everywhere, leaving a model to train on constant features and labels of a single class. The pipeline therefore stops on inputs of that kind, halting if any feature comes out constant, if the labels hold a single class, if a radar event returns no imagery, or if a zonal summary overlaps no

unit. Units without mapped assets are left unscored instead of scored zero, and a missing output is reported as missing.

Each of these checks, and the clipping of every region to the country, is guarded by a test in a suite of 169, which also fails if the dashboard shows mock data or random values. No figure in the documentation is typed by hand, because a build script reads each region's outputs and writes one LaTeX macro per value. A value the pipeline has not produced renders as a visible marker, and a test checks that every macro used in the documentation exists. Rerunning the pipeline and rebuilding a document therefore updates its text and its results together.

3. Illustrative examples

The pipeline was developed on three bounding-box regions, Rangpur and Rajshahi for riverine flooding, Sylhet for flash flooding and the south-west coast for cyclone surge, and then pointed at the whole country. The national partition assigns each of the 61 lowland districts to one of 8 regions by flood regime, and each region needed only a configuration file listing its districts and its flood events, every event checked against a situation report and against Sentinel-1 coverage of its window before the run. The same commands scored 114,569 mapped assets (Fig. 3), and the web build placed them in the national administrative hierarchy (Fig. 2).

The figure carries the evidence a user needs before trusting a region. Training on radar-observed flooding beats training on terrain thresholds in every region, though in West-central only by a margin within noise. The graph network the system was first built around trails the best ordinary model in every region, by 0.005 to 0.090 in AUC, which is why gradient boosting is the default, and the comparison stays in the pipeline so that a future model has to beat it on the same blocks.

Table 2 sets the default model's held-out skill beside the references it has to beat. Its AUC of 0.724 to 0.914 compares with 0.49 to 0.66 for ranking assets by the wetness index alone, the terrain measure on which threshold maps rest. Precision, recall and F1 are low beside published Bangladesh studies because floods make up only 3 to 14 % of held-out assets, whereas those studies score balanced samples of flood and non-flood points. An F1 of 0.12 to 0.44 is read against the 0.05 to 0.24 of the wetness index on the same blocks. Every figure carries its spread over the block assignments, which is

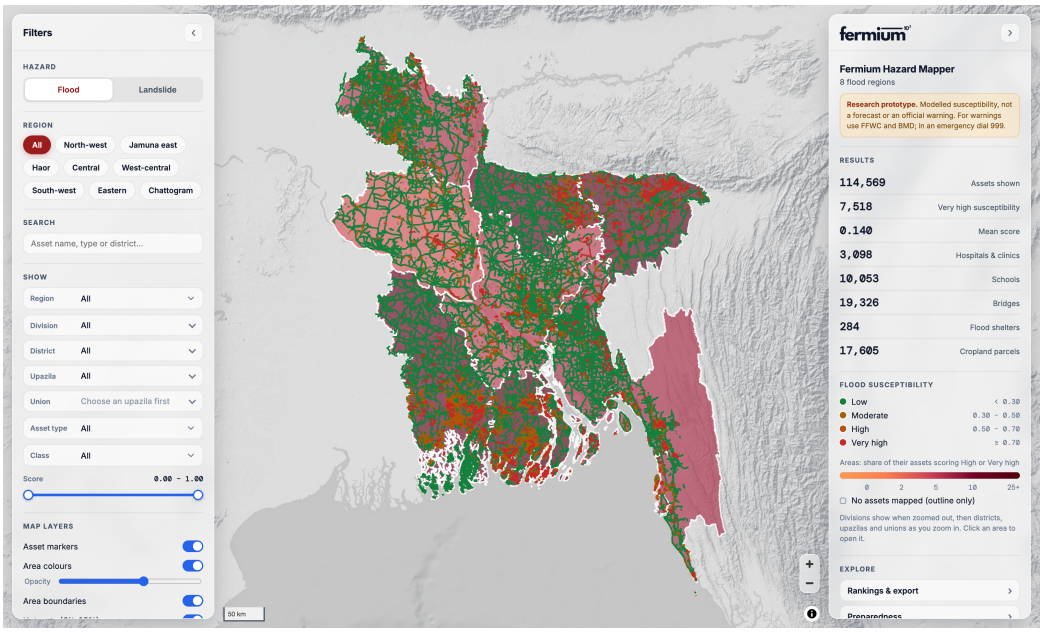


Figure 2: The web application in its national flood view. Asset markers are coloured by modelled susceptibility class and divisions by the share of their assets in the High or Very high class; zooming in replaces divisions with districts, upazilas and unions. Filters, administrative pickers and layers sit on the left; counters for whatever the filters leave visible, the legend and the model figures sit on the right.

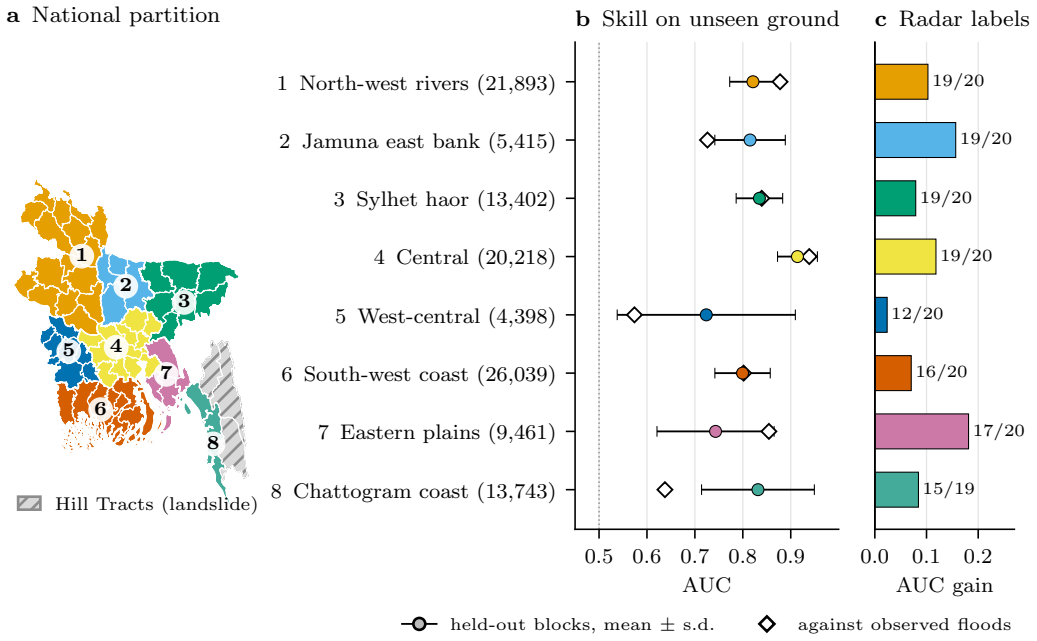


Figure 3: The national partition. (a) The 8 flood regions, numbered as in (b) and (c), and the Hill Tracts landslide region. (b) AUC of the gradient-boosted model on held-out 10 km blocks, as mean and standard deviation over 20 block assignments, and on the held-out blocks of one assignment scored against observed floods; mapped assets in brackets. (c) AUC won by training on radar-observed flooding instead of terrain-threshold labels, with the number of assignments in which radar labels came out ahead.

the uncertainty of the score itself, and the spread of AUC is widest in West-central, the Eastern plains and the Chattogram coast, the regions whose mapped events fall in one or two years. The random forest scores above the default in 8 regions, by 0.001 to 0.085 in AUC, but within one standard deviation in 8 of them, so gradient boosting stays the default and results remain comparable with the archived release.

Vegetation, soil and climate layers are available as optional features, and the covariate test measures whether they earn a place. Adding dry-season NDVI from MODIS, topsoil clay from SoilGrids and mean annual rainfall from CHIRPS raised held-out AUC by 0.013 to 0.076, significantly in 4 regions, while adding only the projected coordinates of each asset raised it by 0.008 to 0.059, 74 % of the layers' gain summed over the regions. Much of what the layers add therefore comes from telling the model where an asset

		Default model (gradient boosting)				RF	Wetness index		ΔAUC when added	
Region	Floods (%)	AUC	Precision	Recall	F1	AUC	AUC	F1	Covariates	Coordinates
1 North-west rivers	6.7	0.821 ± 0.049	0.205	0.385	0.229 ± 0.094	0.822	0.513	0.097	+0.038	+0.021
2 Jamuna east bank	7.3	0.815 ± 0.074	0.263	0.463	0.269 ± 0.119	0.828	0.512	0.098	+0.021	+0.033
3 Sylhet haor	13.5	0.835 ± 0.049	0.343	0.683	0.439 ± 0.091	0.839	0.557	0.238	+0.030	+0.025
4 Central	3.2	0.914 ± 0.042	0.278	0.388	0.261 ± 0.230	0.918	0.567	0.047	+0.013	+0.020
5 West-central	3.1	0.724 ± 0.186	0.145	0.297	0.160 ± 0.115	0.809	0.658	0.073	+0.038	+0.022
6 South-west coast	5.5	0.799 ± 0.058	0.149	0.486	0.212 ± 0.085	0.810	0.557	0.105	+0.063	+0.033
7 Eastern plains	5.2	0.743 ± 0.122	0.112	0.442	0.159 ± 0.088	0.754	0.494	0.088	+0.076	+0.059
8 Chattogram coast	4.4	0.832 ± 0.118	0.110	0.303	0.121 ± 0.089	0.845	0.513	0.064	+0.021	+0.008

Table 2: Held-out skill in each flood region, as the mean over 20 assignments of 10 km test blocks, with the standard deviation after ±. Precision, recall and F1 are taken at the score threshold that maximises F1 on calibration blocks, which neither training nor testing uses. Floods is the share of test assets labelled flooded. RF is a random forest on the same blocks, and the wetness index ranks assets by the topographic wetness index alone. The last two columns give the change in AUC when dry-season NDVI, topsoil clay and mean annual rainfall, or the projected coordinates of each asset alone, are added to the default features.

lies, which helps between neighbouring test blocks without describing the ground, although they beat the coordinates in 6 of 8 regions. Dry-season NDVI is the largest single contributor in 5 regions and rainfall, smooth at 5 km and so close to a coordinate, in the other 3, while clay adds at most 0.027 and lowers AUC in 4 regions. The layers stay optional, since they would make Earth Engine a requirement of every stage, not only of the radar stage, and a fixed default keeps results comparable with the archived release.

The temporal test compares the model with the flood record itself. In the riverine validation regions, a model trained only on floods up to 2022 ranks the ground the 2024 floods reached at 0.801 and 0.797, and the radar record of earlier flooding ranks it at 0.896 and 0.851.

4. Impact

Flood susceptibility studies for Bangladesh report AUCs between 0.81 and 0.984 from random splits of their own points (Section 1), a design that scores a model mostly beside its training data and gives one number without a spread. The software replaces that number with a validation design in which every comparison runs on the same 10 km blocks held out from training, repeated over 20 block assignments and reported with its spread, so a claim about a model or a data source becomes a figure that another group can check with one command.

That design opens questions earlier maps could not answer. Whether a model knows more than the flood record is tested by training on floods up to a cut-off year and scoring the next floods against the radar record of earlier flooding, a baseline that needs no model, and in the riverine validation regions the record ranked the 2024 floods better than the model did. Whether radar-observed floods make better training labels than terrain thresholds is tested by training the same model on both over the same blocks, and radar labels came out ahead in every region of the national partition, by margins that range from within noise in West-central to the largest in the Eastern plains (Fig. 3). Whether a new model architecture adds anything is tested by adding it as another configurable type, which places it in the same comparisons, and the graph network the system was first built around added nothing.

The software also reports measures that a susceptibility map alone does not. Each asset carries a calibrated flood probability, fitted on blocks the model never trained on, beside its ranking score. Each administrative unit from the union to the division is scored by the share of its mapped assets in the High or Very high class, summed over every flood region, so a division spanning several regions is scored from all of them and a unit without mapped assets is reported as missing rather than as safe. Composite risk is written once for infrastructure and once for people, so the two readings of exposure can be set side by side. These figures are served openly at https://fermium.systems/hazmapper for the whole country, beside the official scenario maps.

Reproducibility is built into the software rather than promised. No number in its documentation is typed by hand, one command reruns a validation region from the data archive without Earth Engine and checks every headline figure against it, and the pipeline stops on constant features, labels of a single class or radar events without imagery instead of publishing their results. The pipeline is not tied to Bangladesh's regions, since a new region needs only a configuration file with a bounding box or a list of districts, a country code, a projected coordinate system and a list of flood events with their dates. All inputs come from global open datasets, the Sentinel-1 stage and the optional covariates need only a noncommercial Earth Engine project, and the event lists and asset weights that remain specific to Bangladesh are set in configuration files, not in the code. The input data and outputs of the three validation regions and the landslide region are archived with checksums and licences [24].

5. Discussion and limitations

The held-out AUCs sit below the 0.81 to 0.984 reported for Bangladesh from random splits, and part of that difference is expected from the design alone, since a random split scores a model mostly beside its own training points [5, 6]. The same design carries to other regions and hazards, because a region enters through a configuration file and every input except the radar stage is global, but its scores do not travel with it. A new region has to be validated on its own held-out blocks, and the wide spreads of the regions with one or two flood years show how much a short record limits what any region can claim.

The validation the software runs also marks what its results cannot support. Where a radar record of past floods exists, it ranked the next riverine floods better than the model, so the maps complement that record rather than replace it. The terrain hazard surface is close to chance on the southwest coast, at 0.542 across the grid, and near or below chance wherever embankments, tides or flash floods decide where water lies, as on the Jamuna east bank, because terrain does not describe them. Regions with few mapped floods give unstable scores, and West-central, whose two waterlogging events were documented only in newspapers, varies widely between block assignments and is marked as low-data on the dashboard.

The labels cover the radar era only. The 24 event windows mapped run from 2017 to 2025, so floods before Sentinel-1, such as those of 1988 and 1998, cannot enter them, and the maps describe susceptibility to the floods of the past decade rather than to rarer extremes. The composites draw on every VV scene in their windows, from whichever relative orbit, which mixes incidence angles in exchange for catching each flood peak.

Two further limits sit in the inputs rather than the models. Exposure depends on OpenStreetMap, whose completeness varies widely [25] and has not been measured for rural Bangladesh, so composite risk describes mapped exposure. Vulnerability uses population density and three access distances and leaves out income, housing and age structure. The dashboard states each of these limits beside the map it qualifies.

6. Conclusions

Fermium Hazard Mapper joins radar-observed flood labels, spatially blocked validation and a public dashboard in one open and reproducible

chain, now covering the whole of Bangladesh. Its main contribution is a validation suite that runs every comparison on identical held-out areas and reports it with its spread, together with a design that stops on silent errors instead of publishing their results. Across the national partition that suite places the default model well ahead of the wetness index on unseen ground, finds radar-observed labels ahead of terrain thresholds in every region, and shows vegetation, soil and rainfall layers adding little beyond what location alone supplies.

The next versions follow from what the validation shows. Embankment and tide data would give the surge and flash-flood regions a hazard surface that terrain alone cannot, more mapped events would firm up low-data regions such as West-central, and retraining the landslide module on the multi-year inventory of Rabby and Li [26] would extend the same validation to a second hazard.

CRediT authorship contribution statement

Md Rakib Hasan: Conceptualization, Methodology, Software, Validation, Formal analysis, Data curation, Visualization, Writing – original draft, Writing – review & editing. **Shoumik Zubyer:** Conceptualization, Writing – review & editing.

Declaration of competing interest

Md Rakib Hasan is affiliated with Fermium Systems, which hosts the public dashboard. The software is released under the MIT licence and the dashboard is free to use. The authors declare no other competing interests.

Data availability

The code is at https://github.com/rakibhhridoy/rimes-mapathon, and the release this article describes is archived at https://doi.org/10.5281/zenodo.23108192 [27]. Input data and outputs for the three validation regions and the landslide region are archived at https://doi.org/10.5281/zenodo.23108572 [24], and the national partition is reproduced by the configuration files in the repository. A single command, python scripts/demo.py, downloads a validation region's archive, reruns the pipeline on it without Earth Engine in a few minutes, and checks every headline figure against the archived outputs.

Acknowledgements

The authors thank the Regional Integrated Multi-Hazard Early Warning System for Africa and Asia (RIMES) for organising the Mapathon in which the first version of the software competed. Infrastructure data are © OpenStreetMap contributors, under the Open Database Licence. Population data are from WorldPop, University of Southampton, CC BY 4.0. Administrative boundaries are from geoBoundaries, William & Mary geoLab, CC BY 4.0. Sentinel-1 data are Copernicus Sentinel data, processed in Google Earth Engine. Landslide locations are from NASA's Cooperative Open Online Landslide Repository. The scenario maps shown beside the results are the Department of Disaster Management's Multi-hazard Risk and Vulnerability Assessment, served by the United Nations Satellite Centre (UNOSAT), and the basemap relief is from Esri.

References

[1] A. R. M. T. Islam, S. Talukdar, S. Mahato, S. Kundu, K. U. Eibek, Q. B. Pham, A. Kuriqi, N. T. T. Linh, Flood susceptibility modelling using advanced ensemble machine learning models, Geoscience Frontiers 12 (2021) 101075. doi:10.1016/j.gsf.2020.09.006.

[2] S. Talukdar, B. Ghose, Shahfahad, R. Salam, S. Mahato, Q. B. Pham, N. T. T. Linh, R. Costache, M. Avand, Flood susceptibility modeling in Teesta River basin, Bangladesh using novel ensembles of bagging algorithms, Stochastic Environmental Research and Risk Assessment 34 (2020) 2277–2300. doi:10.1007/s00477-020-01862-5.

[3] R. Islam, P. Chowdhury, Local-scale flash flood susceptibility assessment in northeastern Bangladesh using machine learning algorithms, Environmental Challenges 14 (2024) 100833. doi:10.1016/j.envc.2023.100833.

[4] M. S. G. Adnan, Z. S. Siam, I. Kabir, Z. Kabir, M. R. Ahmed, Q. K. Hassan, R. M. Rahman, A. Dewan, A novel framework for addressing uncertainties in machine learning-based geospatial approaches for flood prediction, Journal of Environmental Management 326 (2023) 116813. doi:10.1016/j.jenvman.2022.116813.

[5] D. R. Roberts, V. Bahn, S. Ciuti, M. S. Boyce, J. Elith, G. Guillera-Arroita, S. Hauenstein, J. J. Lahoz-Monfort, B. Schröder, W. Thuiller,

D. I. Warton, B. A. Wintle, F. Hartig, C. F. Dormann, Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure, Ecography 40 (2017) 913–929. doi:10.1111/ecog.02881.

[6] P. Ploton, F. Mortier, M. Réjou-Méchain, N. Barbier, N. Picard, V. Rossi, C. Dormann, G. Cornu, G. Viennois, N. Bayol, A. Lyapustin, S. Gourlet-Fleury, R. Pélissier, Spatial validation reveals poor predictive performance of large-scale ecological mapping models, Nature Communications 11 (2020) 4540. doi:10.1038/s41467-020-18321-y.

[7] A. Twele, W. Cao, S. Plank, S. Martinis, Sentinel-1-based flood mapping: a fully automated processing chain, International Journal of Remote Sensing 37 (2016) 2990–3004. doi:10.1080/01431161.2016.1192304.

[8] K. Uddin, M. A. Matin, F. J. Meyer, Operational Flood Mapping Using Multi-Temporal Sentinel-1 SAR Images: A Case Study from Bangladesh, Remote Sensing 11 (2019) 1581. doi:10.3390/rs11131581.

[9] A. Haghizadeh, R. Fathiganji, E. Sohrabi, A. Lotfi, L. Ghasemi, A framework for flood risk zoning and prioritization combining maximum entropy and game theory, Scientific Reports 15 (2025) 24153. doi:10.1038/s41598-025-08220-x.

[10] H. Mirzapour, A. Haghizadeh, M. S. Motlagh, Evaluating machine learning efficiency and accuracy for real time flash flood mapping, Scientific Reports 16 (2025) 3975. doi:10.1038/s41598-025-34037-9.

[11] R. Valavi, J. Elith, J. J. Lahoz-Monfort, G. Guillera-Arroita, blockCV: An R package for generating spatially or environmentally separated folds for k-fold cross-validation of species distribution models, Methods in Ecology and Evolution 10 (2019) 225–232. doi:10.1111/2041-210X.13107.

[12] H. Meyer, E. Pebesma, Predicting into unknown space? Estimating the area of applicability of spatial prediction models, Methods in Ecology and Evolution 12 (2021) 1620–1633. doi:10.1111/2041-210x.13650.

[13] G. Nearing, D. Cohen, V. Dube, M. Gauch, O. Gilon, S. Harrigan, A. Hassidim, D. Klotz, F. Kratzert, A. Metzger, S. Nevo, F. Pappenberger, C. Prudhomme, G. Shalev, S. Shenzis, T. Y. Tekalign,

D. Weitzner, Y. Matias, Global prediction of extreme floods in ungauged watersheds, Nature 627 (2024) 559–563. doi:10.1038/s41586-024-07145-1.

[14] P. Salamon, N. McCormick, C. Reimer, T. Clarke, B. Bauer-Marschallinger, W. Wagner, S. Martinis, C. Chow, C. Böhnke, P. Matgen, M. Chini, R. Hostache, L. Molini, E. Fiori, A. Walli, The new, systematic global flood monitoring product of the Copernicus Emergency Management Service, in: 2021 IEEE International Geoscience and Remote Sensing Symposium (IGARSS), 2021, pp. 1053–1056. doi:10.1109/IGARSS47720.2021.9554214.

[15] Department of Disaster Management, Bangladesh hazard & risk map: Multi-hazard risk and vulnerability assessment, Web application by the United Nations Satellite Centre (UNOSAT), https://unosat-geodrr.cern.ch/apps/BGD/hazard-risk/, 2026. Accessed 2 October 2026.

[16] UN-SPIDER, Recommended Practice: Flood Mapping and Damage Assessment Using Sentinel-1 SAR Data in Google Earth Engine, https://un-spider.org/advisory-support/recommended-practices/recommended-practice-google-earth-engine-flood-mapping, 2019.

[17] J.-F. Pekel, A. Cottam, N. Gorelick, A. S. Belward, High-resolution mapping of global surface water and its long-term changes, Nature 540 (2016) 418–422. doi:10.1038/nature20584.

[18] W. L. Hamilton, R. Ying, J. Leskovec, Inductive Representation Learning on Large Graphs, Advances in Neural Information Processing Systems 30; arXiv:1706.02216, 2017. arXiv:1706.02216.

[19] B. Zadrozny, C. Elkan, Transforming classifier scores into accurate multiclass probability estimates, in: Proceedings of the eighth ACM SIGKDD international conference on Knowledge discovery and data mining, 2002, pp. 694–699. doi:10.1145/775047.775151.

[20] European Commission Joint Research Centre, Index for Risk Management – INFORM: Concept and Methodology, Version 2017, Technical Report, Publications Office of the European Union, 2017. doi:10.2760/094023.

[21] A. Getis, J. K. Ord, The Analysis of Spatial Association by Use of Distance Statistics, Geographical Analysis 24 (1992) 189–206. doi:10.1111/j.1538-4632.1992.tb00261.x.

[22] M. Barbet-Massin, F. Jiguet, C. H. Albert, W. Thuiller, Selecting pseudo-absences for species distribution models: how, where and how many?, Methods in Ecology and Evolution 3 (2012) 327–338. doi:10.1111/j.2041-210x.2011.00172.x.

[23] B. Tellman, J. A. Sullivan, C. Kuhn, A. J. Kettner, C. S. Doyle, G. R. Brakenridge, T. A. Erickson, D. A. Slayback, Satellite imaging reveals increased proportion of population exposed to floods, Nature 596 (2021) 80–86. doi:10.1038/s41586-021-03695-w.

[24] M. R. Hasan, Fermium hazard mapper: input data and outputs for flood and landslide susceptibility in Bangladesh (version 3), 2026. doi:10.5281/zenodo.23108572.

[25] B. Herfort, S. Lautenbach, J. Porto de Albuquerque, J. Anderson, A. Zipf, A spatio-temporal analysis investigating completeness and inequalities of global urban building data in OpenStreetMap, Nature Communications 14 (2023) 3985. doi:10.1038/s41467-023-39698-6.

[26] Y. W. Rabby, Y. Li, Landslide Inventory (2001–2017) of Chittagong Hilly Areas, Bangladesh, Data 5 (2019) 4. doi:10.3390/data5010004.

[27] M. R. Hasan, Fermium hazard mapper, 2026. URL: https://github.com/rakibhhridoy/rimes-mapathon. doi:10.5281/zenodo.23108192.